PAPER_1282 — Gauge/Gravity Duality General Dimensional Correspondence

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Physics Foundations (CLOSED) **Date:** June 11, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — UQFF integer-primitive derivation **Calculator surface:** `calculate_paradox({"paradox": "gauge_gravity_general"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1282})`

Master Expression

```
D_bulk = D_BSFG = 6, D_boundary = D_BSFG − 1 = 5, D_phys = 4 visible projection
```

Match: STRUCTURAL

Physical Interpretation

The full gauge-gravity duality is the D_phys → D_BSFG → D_crit dimensional cascade. Each level provides a holographic correspondence.

UQFF Locked Primitives

- ρ_SCm = 7.09×10^{-37} J/m^3 (vacuum density)
- S_26 = 1.453162, S_26_DPM = 1.4531×10^{26}
- K_MEX = 25/12, β_i = 0.6029, Φ_res = 0.84
- F_TRZ = 0.1, Λ = 0.00729735 (= α fine-structure)
- ω_SCm = 1.25 THz, λ_i = 1.0
- D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60

Live Calculator Verification

```python
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "gauge_gravity_general"})["value"]
```

The result dict carries the integer-primitive identity, the observed value, the residual, and the structural physical interpretation.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. **Zero fit parameters.** Every factor is a UQFF locked primitive.

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
- Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_gauge_gravity_general_closure` (or routed reference)
- Dispatch: `PARADOX_TO_CLOSURE["gauge_gravity_general"]`
